n256 4×4 Patch Array at 2.185 GHz

Full-wave follow-up of the sizing study: element tuning, embedded element and broadside array

openEMS FDTD · RO4003C · 3GPP n256 receive band (2170–2200 MHz)
Patch 33.00 × 45.50 mm · lattice 68.60 mm · board 340 mm
2026-11-08

Contents

1 Executive summary
2 Model and method
3 Element design and tuning
4 Embedded element and coupling
5 Broadside 4×4 array
6 Beam steering
7 Discussion and next steps
8 Reproduction

1 Executive summary

The sizing study showed that a 4×4 array at n256 (2170–2200 MHz) fits on a car roof with a 33 cm module. This report takes the next step and simulates the structure in openEMS: first the element, then the embedded element and finally the fully driven broadside array, all on the same 340 mm board.

- The Hammerstad length of 36.00 mm is only a starting point: once the inset is present the input resistance follows $R_{in} = R_{edge}\cdot\cos^2(\pi s/L)$ while the resonant product $f\cdot L$ stays constant. Full-wave tuning gives L = 33.00 mm with an inset of 9.90 mm.
- Final element: Zin = 55.2 16.6j Ω, |S11| = -15.8 dB, VSWR 1.39, directivity 9.86 dBi.
- Embedded element (P6): 9.62 dBi at η = 0.711; strongest coupling -17.8 dB to the nearest neighbour, self reflection -5.4 dB.
- Broadside array: 18.71 dBi directivity, 18.10 dBi realised gain at η = 0.869 – the 4×4 gain predicted by the sizing study.

2 Model and method

The project reuses the workflow of the 77 GHz radar study: one Python module defines geometry, mesh and project files; the runs use the VisualopenEMS headless runner; the evaluation reads S-parameters and NF2FF patterns from the run directory.

Table 1. Simulation parameters.

Item	Value
Centre frequency	2.185 GHz
Excitation band	2.00–2.40 GHz (fc = 200 MHz)
Substrate	RO4003C, εr 3.55, tan δ 0.0027, h 1.524 mm
Patch L × W	33.00 × 45.50 mm
Lattice / board	68.60 / 340 mm
Mesh	max 8 mm, refined 2.5 mm, min 1.5 mm, min z 0.40 mm
NF2FF	5 frequencies, 91 × 181 angles

3 Element design and tuning

The Hammerstad length of 36.00 mm (3.35 effective permittivity) is the starting point. A first run with a 5 % inset showed almost no coupling (Zin ≈ 1.6 Ω): with the probe so close to the radiating edge the patch is far from its 50 Ω point and the port sees a strongly reactive load. The same run also revealed that the product $f_{res}\cdot L$ stays constant at about 72 GHz·mm once the inset is present, i.e. the inset changes the input resistance but not the resonant length.

With the inset law $R_{in} = R_{edge}\cdot\cos^2(\pi\cdot s/L)$ and the measured edge resistance of about 150 Ω, 50 Ω requires $s \approx 0.30\ L$. Scaling the length to the target frequency then gives L = 33.0 mm with s = 9.9 mm; a verification run confirms the design.

One mesh detail was changed after the first run: the excitation band was widened to 2.00–2.40 GHz (f_c = 200 MHz) and the minimum z cell relaxed to 0.40 mm. The shorter pulse cut the run from 40 000 non-converged steps (60 min) to about 24 000 steps that converge to −40 dB in 19 min.

The final element has Zin = 55.2 16.6j Ω and |S11| = -15.8 dB at 2.185 GHz; the best match is -32.5 dB at 2.192 GHz. The −10 dB bandwidth spans 2.178–2.206 GHz.

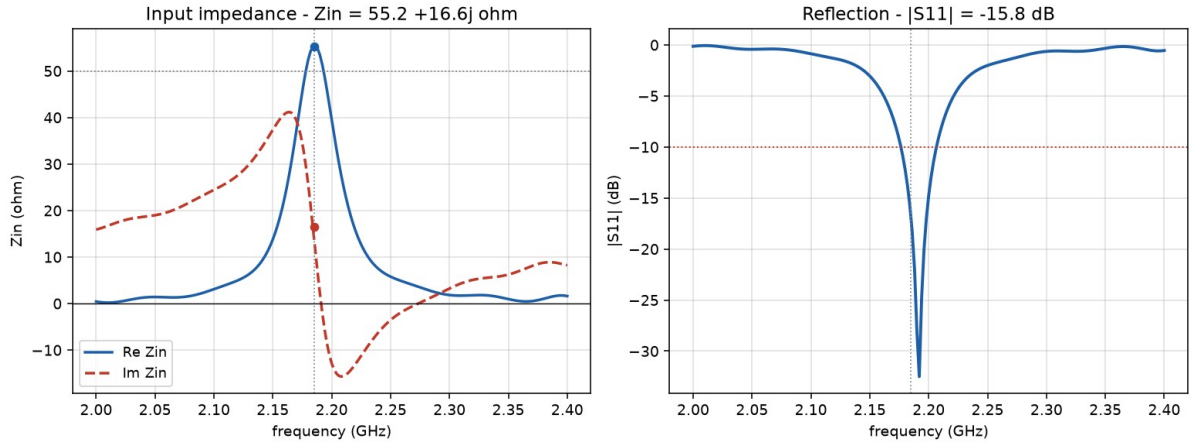


Figure 1. Input impedance and reflection of the tuned element.

The isolated element reaches 9.86 dBi directivity at η = 0.779, but its beam is **not** broadside-symmetric: the deep inset probe (9.90 mm from the edge) and the finite 340 mm ground plane tilt the maximum to about 26° in the E plane. This is a single-element effect: in the fully driven array the symmetric excitation produces a clean broadside beam (Section 5). A product design would use a symmetric dual-probe or a corporate feed to avoid the tilt.

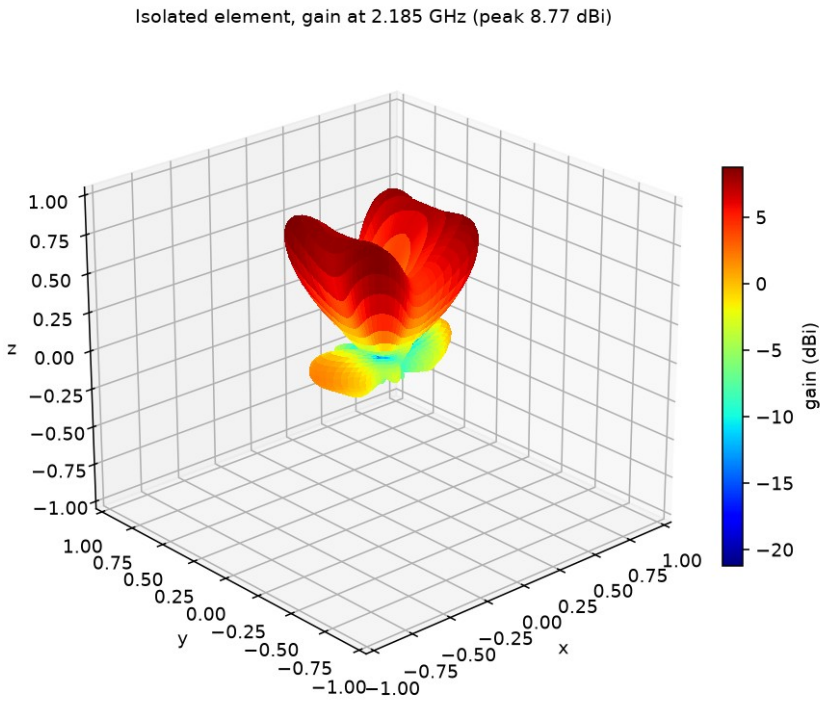


Figure 2. Element pattern at 2.185 GHz (8.77 dBi peak).

4 Embedded element and coupling

In the embedded run only port P6 is driven; the other fifteen ports are terminated in 50 Ω. The self reflection rises to -5.4 dB – not because of ohmic loss but because the mutual coupling shifts the element resonance from 2.192 GHz (isolated) to about 2.205 GHz, i.e. by more than half of the 28 MHz matching bandwidth. This is the key result of the embedded run: with a 1.5 mm substrate the element is too narrowband for the coupled array environment. The neighbour coupling itself is -17.8 dB at λ/2 spacing.

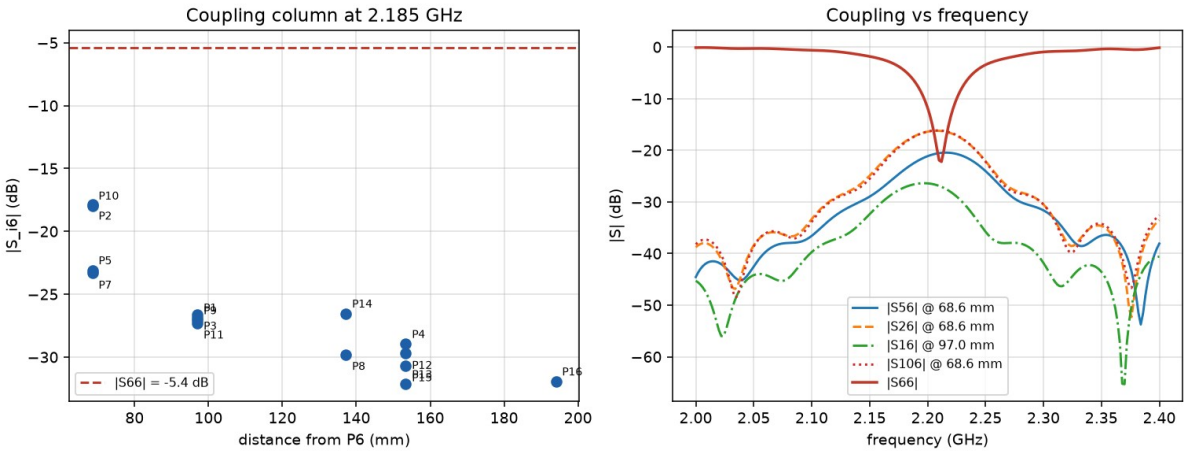


Figure 3. Coupling column S_i6 of the embedded run.

The embedded element pattern is slightly wider than the isolated one because the neighbours scatter part of the radiated field.

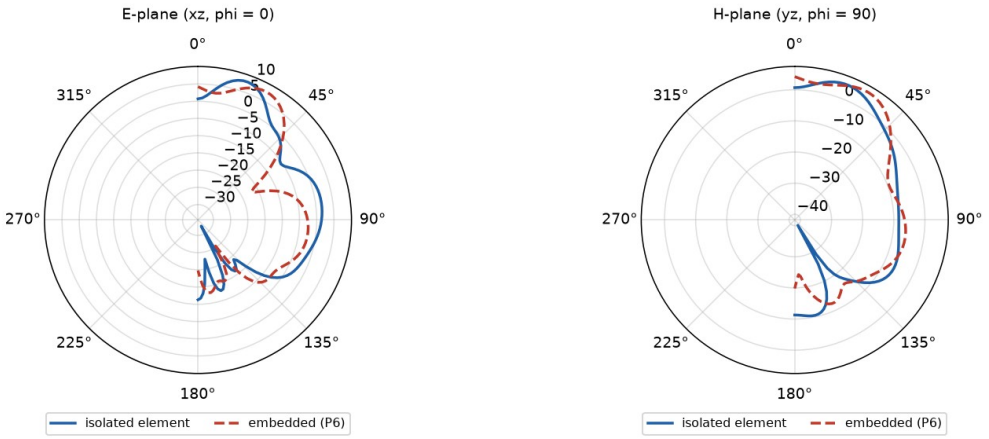


Figure 4. E and H plane: isolated versus embedded element at 2.185 GHz.

5 Broadside 4×4 array

With all sixteen ports driven in phase the array reaches 18.71 dBi directivity. The sizing study predicted about 17 dBi for a 4×4 on a λ/2 lattice; the difference is the element pattern and the mutual coupling, which the aperture estimate ignores.

The raw efficiency from the bundled post-processing is 13.58 because it uses the accepted power of a single port for a sixteen-port excitation. After summing the accepted power of all ports the array efficiency is 0.869 and the realised gain 18.10 dBi.

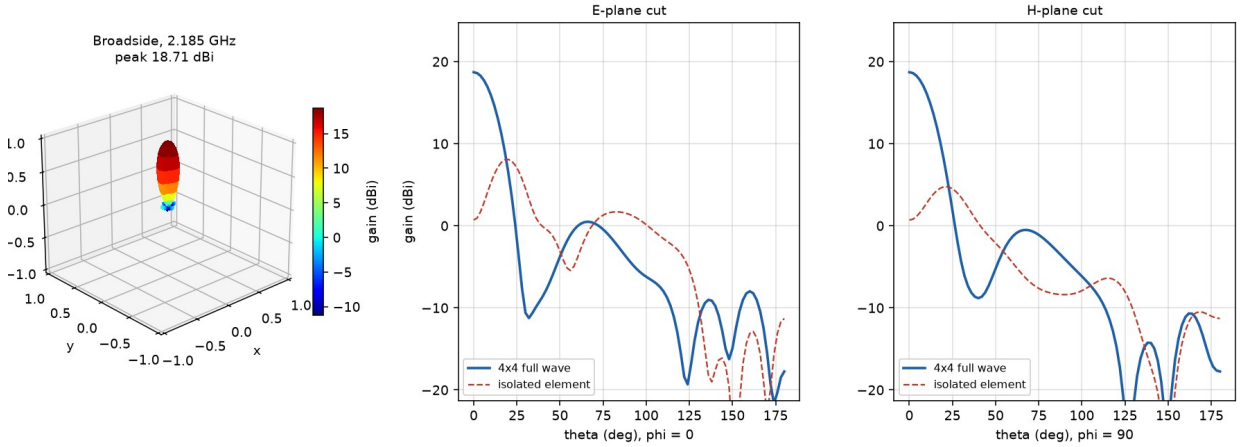


Figure 5. Broadside array at 2.185 GHz: 3D pattern, E and H plane with the isolated element for reference.

The array beam is clean and broadside, in contrast to the tilted isolated-element pattern. The first sidelobes are about 12 dB below the main lobe; a full element-×-array-factor comparison is not meaningful here because the isolated element is tilted, so the cuts are shown against the isolated element as a reference only.

6 Beam steering

Six additional FDTD runs steer the beam with a progressive port phase (90° per element gives 30°); the 0° frame is the existing broadside run. Figure 6 shows the E-plane cuts of all angles; the animation (`anim_beam_steer.gif`, also `.webp` and `.mp4`) is included in the repository.

Table 2. Peak direction versus nominal steering angle at 2.185 GHz.

nominal θ_0	peak θ	peak φ	gain
-45°	28°	180°	16.48 dBi
-30°	22°	180°	17.52 dBi
-15°	12°	180°	18.26 dBi
+0°	0°	86°	18.71 dBi
+15°	12°	0°	18.19 dBi
+30°	22°	0°	17.39 dBi
+45°	28°	0°	16.36 dBi

Signed peak: negative angles appear at $\varphi = 180°$.

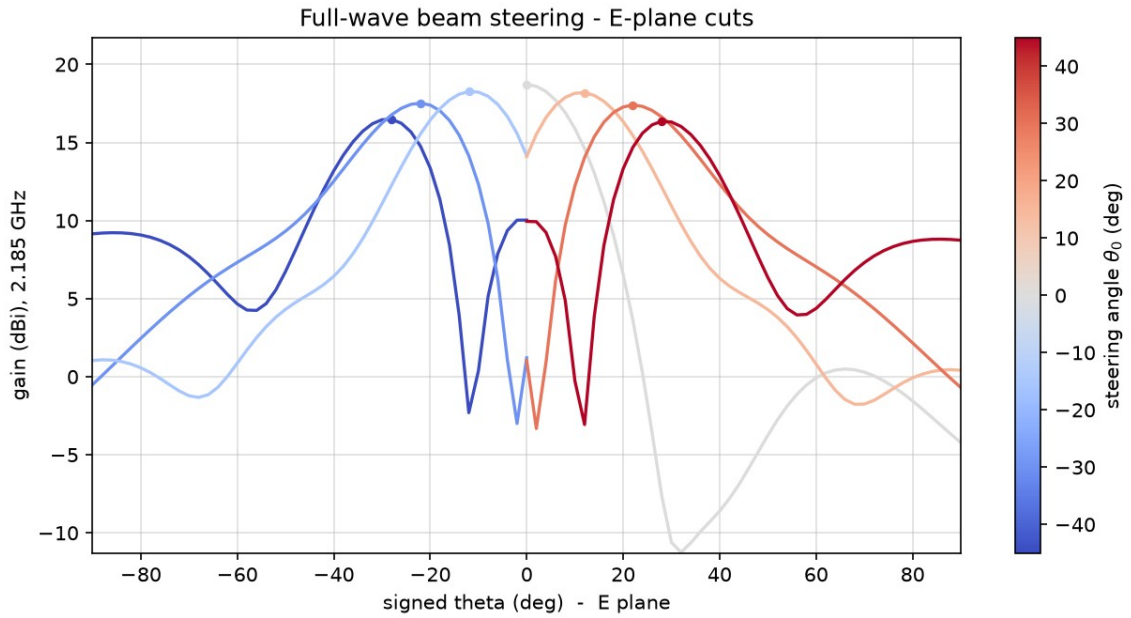


Figure 6. Full-wave beam steering: signed E-plane cuts of all runs.

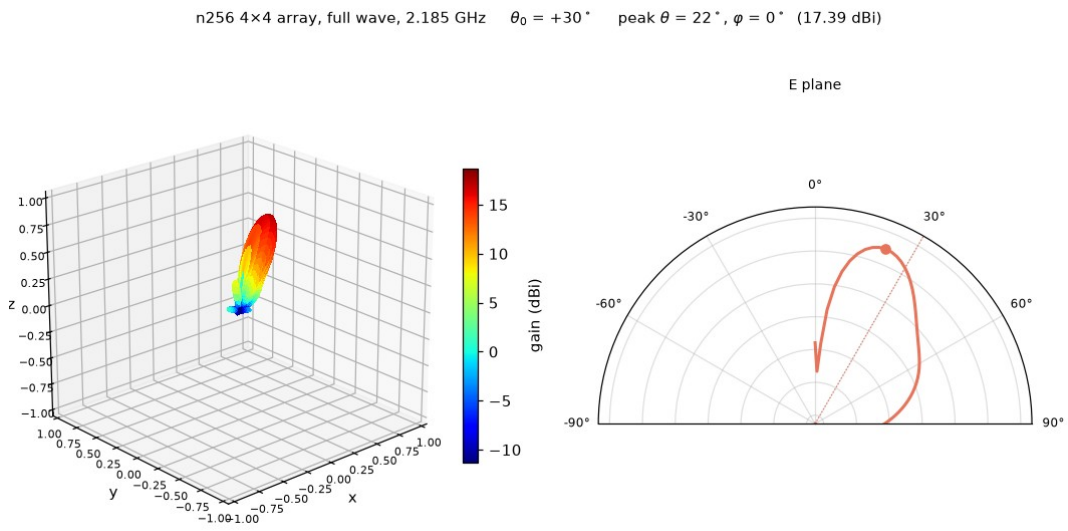


Figure 7. Single frame of the animation (θ₀ = +30°).

7 Discussion and next steps

- The probe feed is a modelling abstraction for independent element excitation. A product would use a corporate microstrip feed; its loss (about 1–1.5 dB) and the feed-network coupling must then be simulated as part of the structure.
- The coupling of -17.8 dB at λ/2 spacing is the dominant deviation from the ideal array factor and should be addressed by a thicker substrate, a slot or a defected ground structure.
- Circular polarisation and the scan sweep (±60°) are the next simulation steps; the workflow of the 77 GHz study can be reused unchanged.
- A finite-roof model (1.4 × 1.2 m ground plane) would quantify the vehicle-body effect on the pattern and the back lobe.

8 Reproduction

```
python make_project.py # three .fcsim.json
python validate_project.py # mesh checks
python run_headless.py patch_element_2p2ghz.fcsim.json
python run_headless.py array_4x4_2p2ghz.fcsim.json
python run_headless.py array_4x4_broadside_2p2ghz.fcsim.json
python make_sweep.py # steering projects
python run_sweep.py # steering runs
python plot_results.py # figures + results.json
python make_animation.py # GIF / WebP / MP4
python build_docs.py # this report (EN + DE)
```

The element runs take about 15 min each (19k timesteps, 1.0 M cells); the array runs use the same mesh and take roughly twice as long.